Power BI Hackathon – Healthcare Operations & Revenue Cycle Challenge

Participant Guide – Group 3

1. Business Context

You are supporting a multi-facility healthcare provider network. Leadership requires a consistent view of clinical operations and revenue cycle performance across four years. The goal is to understand demand patterns, operational pressure, payer performance, and financial risk to improve outcomes and sustainability.

2. Objective

• Explain how demand has changed over time across facilities, departments, and service lines.
• Identify operational pressure points using volume, severity, length of stay, and readmissions.
• Assess financial performance using billed vs allowed vs paid outcomes, and explain the drivers of variation.
• Highlight revenue cycle risks related to denials, delays, and write-offs.
• Provide evidence-based recommendations for leadership actions.

3. Analytical Expectations

• High-quality data preparation and consistent data types across sources
• A clear and efficient data model that supports analysis and filtering
• Accurate calculations aligned to definitions and assumptions
• A KPI layer that covers both clinical and financial performance
• Trend, variance, and segmentation analysis across key metrics
• A coherent story connecting operations to revenue cycle outcomes

4. Deliverables

• A Power BI report with 3–4 pages
• A KPI section with clearly defined measures
• Clinical operations views (volume, severity, LOS, readmissions, satisfaction)
• Revenue cycle views (denials, payment delays, collections, write-offs)
• An executive summary with insights and recommendations supported by analysis

5. Evaluation Criteria

Category	Weight
Data Cleaning & Preparation	25%
Data Modeling Quality	20%
DAX & Calculation Accuracy	20%
Report Structure & Clarity	15%
Depth of Insights & Business Impact	20%

6. Important Note

This challenge reflects a real-world analytics engagement. You are not given a prescribed approach. It is your responsibility to structure, model, and analyze the data appropriately. Results will be evaluated based on analytical quality, clarity of storytelling, and business impact.